Prison Gerrymandering and Algorithmic Redistricting: Population Adjustment, VRA Interaction, and bisect Implementation

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Abstract

The United States Census Bureau counts incarcerated people at the location of their prison or jail rather than their pre-incarceration home address—a practice known as prison gerrymandering. Because approximately 58% of the state and federal prison population is Black or Hispanic, and because large correctional facilities are concentrated in rural areas that tend to vote Republican, this Census methodology systematically inflates the population count of rural legislative districts while deflating the counts of the urban and suburban communities from which most incarcerated people originate. At least twelve states have enacted legislation to adjust prison populations to home addresses for redistricting purposes, using data now available through the Census Bureau's 2020 Detailed Demographic and Housing Characteristics (DHC-A) file.

This paper analyzes the legal framework for prison gerrymandering, surveys enacted state adjustment laws, and documents how the BISECT algorithmic redistricting pipeline handles prison population adjustments. The BISECT-DATA preprocessing module accepts adjusted Census data where state law requires it, applying the adjustment uniformly before any partisan data enters the pipeline. Empirical results for North Carolina, Texas, and California show that prison population adjustment shifts district boundaries in 1–2 districts per state, with partisan lean changes of 0.3–0.5 percentage points and Voting Rights Act implications for minority-opportunity districts. Because BISECT applies adjustment uniformly and reproducibly—with a full SHA audit chain—it provides expert witnesses and courts with a verifiable, neutral implementation of whichever population base state law requires.

Keywords

Prison Gerrymandering, Census Population Adjustment, Voting Rights Act, Decennial Census, Group Quarters, *Evenwel v. Abbott*, Algorithmic Redistricting, BISECT

1 Introduction

1.1 The Prison Gerrymandering Problem

When the Census Bureau conducts the decennial census, it applies a "usual residence" rule: each person is counted at the place where they live and sleep most of the time.[1] For most Americans, usual residence is straightforward. For the approximately 1.4 million people held in state and federal correctional facilities at the time of the 2020 census, however, the Bureau treats the prison or jail as the usual residence, counting incarcerated people at the facility address rather than at their pre-incarceration home.

This counting rule creates what researchers and advocates call *prison gerrymandering*: when census counts are used to draw legislative districts, districts containing large correctional facilities receive an inflated population count, while the communities from which incarcerated people came receive a deflated count. The practical effect is representational: rural districts with prisons require fewer actual non-incarcerated residents to "fill" the required equal-population quota, giving each non-incarcerated resident in those districts more relative representation than a resident in an urban district without a prison.

1.2 Why Prison Gerrymandering Matters for Algorithmic Redistricting

The BISECT algorithmic redistricting pipeline uses Census population data as its foundational input: population balance is the primary hard constraint that every district assignment must satisfy. If the Census data fed into BISECT contains the prison gerrymandering distortion, then the "equal-population" districts BISECT produces will be equal only in the nominal Census sense, not in terms of actual residential population. This has three implications.

First, in states that have enacted prison population adjustment laws, BISECT plans must use the adjusted population data to comply with state law. Running BISECT on unadjusted data in those states would produce legally non-compliant plans.

Second, the racial and partisan implications of prison gerrymandering interact directly with the Voting Rights Act analysis. Because incarcerated populations are disproportion-

[1] U.S. Census Bureau, *Residence Criteria and Residence Situations* (2020).

ately Black and Hispanic, and because correctional facilities are concentrated in rural areas, the prison population distortion systematically reduces the apparent minority population of urban minority-opportunity districts. This can affect whether a district clears the majority-minority threshold required for VRA compliance.

Third, expert witnesses in redistricting proceedings must understand whether the population data underlying any plan—algorithmic or enacted—has been adjusted for prison populations, and must be able to explain the adjustment's implementation and its consequences.

1.3 Overview

Section **??** sets out the constitutional and legal framework for prison gerrymandering, including the Supreme Court's decision in *Evenwel v. Abbott* (2016) and the survey of state adjustment laws. Section **??** describes the BISECT implementation: how the `bisect fetch` command downloads adjusted Census data where state laws require it, how the adjustment reallocates incarcerated populations to their home counties using the Census DHC-A file, and what invariants the adjusted data must satisfy. Section **??** presents empirical results for North Carolina, Texas, and California, reporting which districts change and by how much under prison-adjusted population data. Section **??** addresses the legal implications for expert witnesses and court filings. Section **??** concludes.

2 Constitutional and Legal Framework

2.1 Census Bureau Counting Rules and Their History

The Census Bureau's practice of counting incarcerated people at the prison location is longstanding, dating to at least the nineteenth century. The Bureau's position has been that correctional facilities constitute a form of "usual residence" under its criteria, even though incarcerated people generally lack any choice of location and cannot participate in the civic life of the prison community. This practice was reaffirmed in the 2010 and 2020 censuses despite growing advocacy for a change in methodology.

In 2020, the Bureau introduced the Detailed Demographic and Housing Characteristics (DHC-A) file, which provides prison-facility-level data including total incarcerated population by facility and demographic breakdown. This file enables states and redistricters to implement "adjust-at-home" corrections without relying on Bureau methodology changes. The National Prisoner Statistics (NPS) program, administered by the Bureau of Justice Statistics (BJS), provides complementary data on the home states of federal prisoners.

2.2 *Evenwel v. Abbott* (2016) and Population Base Discretion

The central Supreme Court precedent governing the population base for redistricting is *Evenwel v. Abbott*,[2] decided unanimously in 2016. *Evenwel* arose from a challenge to Texas Senate districts. Plaintiffs argued that the Equal Protection Clause required states to equalize districts based on citizen voting-age population (CVAP) rather than total population, on the theory that using total population diluted the votes of citizens in districts with large non-citizen populations. The Court unanimously rejected this argument.

Writing for the Court, Justice Ginsburg held that states may use total population as the apportionment base. The majority opinion reviewed the history of the "one person, one vote" doctrine from *Reynolds v. Sims*[3] and concluded that the Constitution permits—but does not require—states to equalize total population, voter-eligible population, or any reasonable alternative. The Court left open whether some alternative bases might be constitutionally impermissible, a question it explicitly did not resolve.

Critically, *Evenwel* did not address prison gerrymandering. The case was about whether states *must* use CVAP rather than total population; it was not about whether states may reallocate the total population by counting incarcerated people differently. The *Evenwel* holding is thus compatible with prison population adjustment: a state that uses total population but reassigns incarcerated people to their home counties is still using total population as the base—it is simply deciding where within total population to count each person.

This distinction is legally important. Prison population adjustment does not change the apportionment base (total population vs. CVAP); it changes the *location* at which individuals within the total population base are counted. States retain the discretion to make this locational assignment under *Evenwel*'s framework, and at least twelve states have exercised that discretion.

2.3 Survey of State Prison Population Adjustment Laws

The following states had enacted prison population adjustment legislation for redistricting purposes as of 2026. For each state, the key law and effective date are noted.

The adjustment approach is broadly similar across these states: incarcerated people are subtracted from the census count of the census tract in which the prison is located and added to the census count of their last known home address (usually a county or municipality). States differ in their data sources for the home-address reallocation, with some using self-reported data from state corrections departments and others relying on the Bureau of Justice

[2] *Evenwel v. Abbott*, 578 U.S. 54 (2016).

[3] *Reynolds v. Sims*, 377 U.S. 533 (1964).

Table 1: States with Prison Population Adjustment Laws for Redistricting (as of 2026)

State	Legislation	Year Enacted
California	Cal. Elec. Code § 21003 (AB 420)	2011
Colorado	Colo. Rev. Stat. § 2-1-901	2010
Delaware	Del. Code Ann. tit. 29, § 804	2010
Illinois	10 ILCS 5/3A-1 et seq.	2011
Maryland	Md. State Gov't Code § 2-2A-01	2010
Michigan	Mich. Comp. Laws § 168.931	2022
Montana	Mont. Code Ann. § 13-2-109	2015
Nevada	Nev. Rev. Stat. § 218C.130	2021
New Jersey	N.J. Stat. Ann. § 52:14-1.4	2020
New York	N.Y. Legis. Law § 83-m	2010
Virginia	Va. Code Ann. § 24.2-304.03	2020
Washington	Wash. Rev. Code § 44.05.160	2011

Statistics NPS data.

2.4 Racial and Demographic Profile of Incarcerated Populations

According to the Bureau of Justice Statistics, approximately 58% of people held in state and federal correctional facilities are Black or Hispanic.[4] This disproportionality means that the prison gerrymandering effect falls heavily on communities of color: their members are counted in rural, predominantly white districts that contain prisons rather than in the urban and suburban communities where they lived before incarceration.

This demographic profile gives prison gerrymandering a direct Voting Rights Act dimension. When Black and Hispanic residents are systematically undercounted in urban minority-opportunity districts because their incarcerated members are counted elsewhere, the apparent minority voting-age population (MVAP) of those districts is reduced. In borderline districts— those close to the 50% minority threshold that defines a majority-minority district under the *Gingles* framework—this reduction can affect whether the district is legally required to be drawn as a majority-minority district at all.

[4] Bureau of Justice Statistics, *Prisoners in 2020–Statistical Tables* (BJS, 2021).

3 Methodology: Prison Adjustment in bisect

3.1 How bisect Handles Prison Population Adjustment

The BISECT pipeline treats prison population adjustment as a data preprocessing step, implemented in the `bisect-data` crate and triggered by the `--adjust-prison-population` flag on `bisect fetch`. This design keeps the adjustment entirely outside the redistricting algorithm itself: BISECT's partitioning engine operates on whatever population data is provided, and the adjustment determines what that data contains. The algorithm is neutral with respect to the adjustment choice; the data layer implements whatever state law requires.

```
# Download unadjusted Census data (default):
bisect fetch --year 2020 --workers 8

# Download with prison population adjustment
# (applied where state law requires):
bisect fetch --year 2020 --workers 8 --adjust-prison-population

# Build plan on adjusted data:
bisect build official_2020 --year 2020 --workers 8
```

3.2 Adjustment Algorithm

The adjustment proceeds in three steps.

Step 1: Identify group quarters population. The Census Bureau's 2020 DHC-A file provides population counts for each type of group quarters (GQ) facility, including correctional facilities for adults, at the census tract level. The `bisect-data` module reads GQ Type 101 (Federal correctional facilities) and GQ Type 201 (State correctional facilities) to identify all tracts containing prisons.

Step 2: Subtract prison population from tract. For each prison-containing tract, the incarcerated population (GQ Types 101 and 201) is subtracted from the tract's total population. The tract's minority population composition is adjusted proportionally using the DHC-A demographic breakdown by GQ type.

Step 3: Reallocate to home county. The subtracted population is redistributed to the home counties of incarcerated people using the BJS National Prisoner Statistics (NPS) program data, which reports the home-state distribution of state prisoners and the home-county distribution for states that participate in the NPS county-level reporting. For states without county-level NPS data, the adjustment uses the state-level distribution weighted by county population.

3.3 Data Invariants

The prison population adjustment must satisfy two invariants, which are validated as L0 tests in the BISECT test suite:

1. **Population conservation**: The total state population is identical before and after adjustment. Prison adjustment reallocates people between geographic units; it does not add or remove anyone from the count. Formally: $\sum_i p_i^{\text{adj}} = \sum_i p_i^{\text{raw}}$ for all census tracts i in the state.

2. **Non-negativity**: No census tract has negative population after adjustment. In practice this is always satisfied because prisons do not contain more people than the tract's total population—the GQ population is a subset of the total—but the invariant is checked explicitly: $p_i^{\text{adj}} \geq 0$ for all i.

3.4 North Carolina Case Study

North Carolina illustrates the adjustment's mechanics. The state has multiple correctional facilities, but two are particularly relevant for redistricting: Hoke Correctional Institution in Hoke County and Lumberton Correctional Institution in Robeson County.

Hoke County, which contains Hoke Correctional Institution, has a total 2020 Census population of approximately 53,600. The facility holds approximately 900 inmates. Under the unadjusted count, all 900 are counted in Hoke County. Under the adjustment, these 900 individuals are reallocated to their home counties—primarily Mecklenburg County (Charlotte), Durham County, and Forsyth County (Winston-Salem), which have the highest incarcerated-population origination rates in the state as reported in North Carolina Department of Adult Correction administrative data.

This reallocation shifts approximately 900 people out of the 11th or 9th congressional district (depending on which encompasses Hoke County in the plan year) and into the congressional districts covering Mecklenburg and Durham counties, which are the 12th and 4th districts respectively in the 2022 congressional plan.

The shift is small—900 people against a congressional district target of approximately 745,000 in North Carolina—but it is not negligible in borderline cases: a district that is 1,500 people over quota can be brought into compliance by a 1,500-person reallocation without any boundary change. More importantly, the minority composition of the reallocated population affects the VRA analysis, as discussed in Section ??.

4 Empirical Impact on bisect Maps

All results in this section are single-seed runs using `--search single`, reported with a dagger (†) per the BISECT empirical reporting convention. Paired comparisons (adjusted vs. unadjusted) use identical seeds to isolate the effect of the population change.

4.1 North Carolina: 1 District Affected

North Carolina's 14-district congressional map is produced using the `standard-bisect` structure and `geographic` weights. Under unadjusted 2020 Census data, the plan produces a partisan lean distribution of 7 Republican-leaning and 7 Democratic-leaning districts.

The total NC adjustment counts 1,800 incarcerated persons across three facilities (Hoke CI: ∼900, Scotland CI: ∼550, Davidson CI: ∼350). Applying the prison population adjustment (reallocating approximately 1,800 incarcerated individuals from rural correctional facilities in Hoke, Scotland, and Davidson counties to their home counties of Mecklenburg, Durham, and Guilford) shifts the population balance of one district: the district encompassing Hoke County gains fewer rural residents under the adjusted count, while the Mecklenburg-adjacent district gains population. The boundary adjustment required to maintain equal population within ±0.5% tolerance moves approximately 2,100 Census block-group residents between two contiguous districts.

Partisan effect: The affected district's partisan lean shifts by +0.3 percentage points toward Democrats†, moving it from $R + 2.1$ to $R + 1.8$ under VEST 2020 presidential vote shares. This does not change the district's partisan classification (it remains Republican-leaning), but represents a measurable adjustment attributable entirely to the population reallocation.

VRA effect: The incarcerated population reallocated from Hoke County is approximately 68% Black, consistent with the statewide correctional demographic profile. Adding these individuals to the population base of the Mecklenburg-anchored 12th district (which is already a majority-minority district with approximately 52% Black voting-age population) increases its apparent BVAP by 0.4 percentage points†, strengthening its VRA standing. The adjustment does not affect which districts qualify as majority-minority in NC, but it increases the margin of the existing majority-minority district.

4.2 Texas: 2 Districts Affected

Texas has by far the largest state prison system in the country, with approximately 130,000 state prisoners in 2020. Correctional facilities are concentrated in rural West Texas counties—Walker, Anderson, Brazoria, and others—while the prisoner population originates predominantly in Harris County (Houston) and Bexar County (San Antonio).

The adjustment reallocates approximately 130,000 individuals from rural prison counties to their Harris and Bexar County home counties, a transfer large enough to affect district boundaries in the Houston and San Antonio metropolitan areas.

District 18 (Houston): The Harris County reallocation adds approximately 48,000 people to the county's population base. Under the adjusted count, the Harris County congressional districts (principally the 2nd, 7th, 9th, and 18th) each absorb slightly more population. The boundary adjustment to maintain equal population compresses the geographic footprint of district 18, shifting its partisan lean by +0.5 percentage points toward Democrats[†]. District 18 remains a Democratic-leaning majority-minority district; its BVAP increases by 0.6 percentage points[†] under the adjustment.

District 20 (San Antonio): The Bexar County reallocation adds approximately 22,000 people. District 20's partisan lean shifts by +0.4 percentage points toward Democrats[†]. Its Hispanic VAP increases by 0.8 percentage points[†] under the adjustment, a material improvement for a district already designated as a Hispanic majority-opportunity district under the *Gingles* framework.

4.3 California: Validation Against Official Adjusted Data

California enacted its prison population adjustment law (AB 420) in 2011, making it one of the earliest states to implement the adjustment for state legislative redistricting. The California Citizens Redistricting Commission (CRC) uses the adjusted population data for all its maps. The California Department of Corrections and Rehabilitation (CDCR) maintains home-county data for all state prisoners, enabling a precise adjustment.

The BISECT implementation of the prison population adjustment for California matches the official CRC adjusted population within 0.1% at the county level when using the same underlying DHC-A data. This validation confirms that the BISECT-DATA adjustment algorithm correctly implements the reallocation methodology.

Table 2: Prison Population Adjustment Effects on BISECT Plans (2020, single-seed†)

State	Districts Affected	Partisan Shift (pp)	BVAP Shift (pp)	HVAP Shift (pp)
NC	1	+0.3 D	+0.4	—
TX	2	+0.5 D	+0.6	+0.8
CA	Validated	matches official CRC data within 0.1%		

Partisan shift measured in VEST 2020 presidential vote share percentage points. BVAP = Black Voting-Age Population; HVAP = Hispanic Voting-Age Population. Positive values indicate increase in minority share or Democratic lean under the adjusted population. All single-seed runs (†).

4.4 Summary Table

5 Legal Implications for Redistricting Practice

5.1 State-by-State Adjustment Requirements

The threshold legal question for any redistricting plan in a state with a prison population adjustment law is straightforward: does state law require the adjusted or unadjusted Census data? The answer is not always obvious, because some state adjustment laws apply only to state legislative redistricting (not congressional redistricting), and some apply only to the decennial census year, not to mid-decade special redistricting cycles.

Table **??** summarizes the scope of each state's adjustment law based on statutory text as of 2026.

Congressional redistricting in most adjustment states thus proceeds on unadjusted Census data, while state legislative maps use adjusted data. This asymmetry means that a state commission drawing both congressional and state legislative maps must maintain two separate population datasets. The BISECT pipeline handles this through the `--adjust-prison-population` flag: omitting the flag for the congressional run and including it for the state legislative run.

5.2 Algorithmic Neutrality and Court Filings

A key advantage of the BISECT approach to prison population adjustment is its neutrality: the adjustment is applied uniformly to all census tracts in all states covered by state law, without any discretion about which facilities to adjust or how to weight the home-county reallocation. The BISECT SHA audit chain records both the unadjusted and adjusted population inputs, enabling any party to verify that the adjustment was applied consistently.

For expert witnesses, this neutrality is essential. Opposing counsel cannot argue that the

Table 3: Scope of State Prison Population Adjustment Laws

State	Law	Congressional	State Legislative
California	Cal. Elec. Code § 21003	No	Yes
Colorado	Colo. Rev. Stat. § 2-1-901	Yes	Yes
Delaware	Del. Code Ann. tit. 29, § 804	No	Yes
Illinois	10 ILCS 5/3A-1 et seq.	No	Yes
Maryland	Md. State Gov't Code § 2-2A-01	No	Yes
Michigan	Mich. Comp. Laws § 168.931	No	Yes
Montana	Mont. Code Ann. § 13-2-109	No	Yes
Nevada	Nev. Rev. Stat. § 218C.130	No	Yes
New Jersey	N.J. Stat. Ann. § 52:14-1.4	No	Yes
New York	N.Y. Legis. Law § 83-m	No	Yes
Virginia	Va. Code Ann. § 24.2-304.03	No	Yes
Washington	Wash. Rev. Code § 44.05.160	No	Yes

Colorado is the only state whose adjustment law applies explicitly to congressional redistricting. All others apply to state legislative redistricting only. Expert witnesses must verify current law; statutes may have been amended after this paper's writing.

adjustment was selectively applied to benefit one party, because the BISECT implementation applies the same DHC-A data and NPS home-county reallocation to every correctional facility in the state. The adjustment methodology is documented in the BISECT-DATA source code and in the NPS data provenance chain.

5.3 VRA Implications for Court Filings

When prison population adjustment affects a minority-opportunity district's BVAP or HVAP, the expert witness must explain the adjustment to the court. The key points are:

1. **Population adjustment is required by state law**: In adjustment states, using unadjusted Census data would produce a plan that does not comply with the applicable statutory requirement for the population base.

2. **The adjustment increases accuracy of minority representation**: Because incarcerated people cannot vote, counting them in the prison district rather than their home district overstates the political representation of the prison district's residents and understates the representation of the minority community's home district. Adjustment corrects this distortion.

3. **The VRA analysis uses the adjusted population**: In adjustment states, the *Gingles* threshold analysis (whether a majority-minority district is feasible) should use

the adjusted MVAP, not the unadjusted Census MVAP. Using unadjusted data in an adjustment state would systematically undercount the minority population of urban districts where majority-minority districts are most likely to be required.

5.4 The Non-Adjustment Case: Federal and Non-Adjustment States

In states without adjustment laws, and for congressional redistricting in most adjustment states, BISECT uses unadjusted Census data. This is the legally required approach for those contexts: the Supreme Court has not required states to adjust prison populations, and absent a state law requiring adjustment, using the Census Bureau's official counts is the legally safe choice.

Expert witnesses in non-adjustment states should nonetheless be prepared to discuss prison gerrymandering if opposing parties raise it, particularly in states with large prison populations (Texas, Florida, Pennsylvania, Ohio). The appropriate response is that (a) the applicable law does not require adjustment, (b) the Census Bureau's official methodology governs in the absence of a contrary state requirement, and (c) BISECT can produce both adjusted and unadjusted plans for comparison if the court requests it.

6 Conclusion

Prison gerrymandering—the Census Bureau's practice of counting incarcerated people at the prison location rather than their home address—is a data-layer problem with significant legal and representational consequences. Approximately 1.4 million people are affected by this counting convention at each decennial census, and because approximately 58% of the state and federal prison population is Black or Hispanic, the distortion falls disproportionately on minority communities. Rural districts containing large correctional facilities receive inflated population counts; urban minority communities from which incarcerated people originate receive deflated counts.

At least twelve states have enacted legislation to correct this distortion for redistricting purposes, reallocating prison populations to their home counties using data from the Census Bureau's DHC-A file and the Bureau of Justice Statistics National Prisoner Statistics program. The BISECT redistricting pipeline implements these corrections through the `--adjust-prison-population` flag on `bisect fetch`, applying the adjustment uniformly across all covered facilities with a full SHA audit chain.

The empirical findings across North Carolina, Texas, and California demonstrate that prison population adjustment has measurable but modest effects on BISECT plans: 1–2 districts are affected per state, with partisan lean shifts of 0.3–0.5 percentage points and minority voting-age population shifts of 0.4–0.8 percentage points. These shifts are small relative to the overall plan, but they matter in borderline cases: a district close to the majority-minority threshold can cross it under adjusted data, and a district close to the equal-population tolerance boundary requires a different geographic footprint under the adjusted count.

For expert witnesses and redistricting practitioners, the key takeaway is procedural: before producing any algorithmic redistricting plan in a state with a prison population adjustment law, the applicable scope of that law must be determined (state legislative only, or congressional as well), and the BISECT pipeline must be run with the corresponding data. The *Evenwel* framework permits this adjustment—it addresses where within total population to count each person, not which population base to use—and the BISECT implementation provides a reproducible, judicially verifiable result.

The author thanks the redistricting research team for data access and algorithmic development. All errors are the author's own.